

Homework 1 — Marketing Mix Modeling for GLUCOREN

From spend to a defensible reallocation memo

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Released: Week 3 (Monday, September 28, 2026) · **Due:** Monday, October 12, 2026, 11:59 PM ET

Weight: 10% of the final grade · Bonus available (+10 points)

Deliverable: one zip with (a) a Jupyter notebook and (b) a two-page memo as PDF

Data: glucoren_weekly_mmm.csv and glucoren_geo_lift_dtc_tv.csv (the Week 3 class data, in the course folder)

AI policy: permitted with disclosure per the syllabus. The memo must be your own writing.

Scenario

You have joined the commercial analytics team supporting GLUCOREN, the oral primary-care brand from Weeks 1–3, three years after launch. The Brand Lead is preparing the annual commercial plan and has asked the question from class: “We have spent \$78M promoting the brand since launch — about \$26M a year. Which \$10M would you cut, and how sure are you?”

You have 156 weeks of national data: projected TRx, weekly spend in five channels (HCP detailing, conferences, digital, email, DTC TV), the rep call count, a competitor-launch flag (week 90) and the share of commercial lives with preferred formulary status (which steps up at week 118). You also have the results of a DTC-TV geo holdout run on flight 7. The Brand Lead is sharp but not a modeler. She wants three things: a clean decomposition (where did our TRx come from?), defensible ROI per channel with ranges, and a recommended reallocation she can defend to the CFO — with the risks named.

Assumption

Use a contribution of \$250 per TRx when computing ROI (Week 1's GLUCOREN P&L gave $\approx$ \$252). In a real engagement finance gives you this number; never invent it. State the assumption in the memo.

Data

File	Grain	Columns
glucoren_weekly_mmm.csv	156 weekly rows (Mondays, 2023-07-03 → 2026-06-22)	week · trx · hcp_detailing_spend · conferences_spend · digital_spend · email_spend · dtc_tv_spend · hcp_calls · competitor_active (0/1 from week 90) · pbm_preferred_share (0.38 → 0.61 at week 118)
glucoren_geo_lift_dtc_tv.csv	40 DMAs × 16 weeks (8 pre, 8 test)	dma · week · period (pre/test) · group (exposed/holdout) · trx · dtc_tv_spend_alloc

The data is synthetic and was generated from a known response model. The generating parameters are not distributed and are not needed; do not spend time looking for them.

Tasks

Part A — EDA and controls (10 pts)

1. Load the data and produce three plots: weekly TRx with the two events marked; stacked weekly spend by channel; same-week spend vs. TRx per channel with the correlation in the title.
2. Engineer the controls: a logistic launch ramp (choose midpoint and scale by AIC, not by eye alone), an annual sine/cosine pair, the competitor indicator, and the formulary-access variable. Say in one sentence each why every control is there.
3. Show what happens to the channel coefficients when the access control is omitted (refit without it; one small table).

Part B — Adstock and saturation (15 pts)

4. Implement geometric adstock and Hill saturation as two functions. Plot impulse responses for $\delta \in \{0.1, 0.4, 0.7\}$ and Hill curves for three (K, s) pairs; one sentence per parameter.
5. Write down, per channel, the decay range you will allow the search to use and justify each in one line (channel knowledge, not the data). This is your prior.

Part C — OLS MMM and diagnostics (25 pts)

6. Grid-search δ (within your ranges), K (as a multiple of the channel's median adstocked spend) and s per channel by AIC, one channel at a time, at least two sweeps. Report the winning cell and the two runner-up cells per channel with their ΔAIC .
7. Fit OLS with the chosen transforms and the controls. Report coefficients, R^2 , a residual plot with Durbin-Watson, and VIF for every regressor. Comment on signs and any VIF above 5.
8. Hold-out: refit the coefficients on weeks 1–104 and predict 105–156. Report MAPE overall and split at week 118. State plainly what the model could and could not have forecast, and whether your grid was re-run on the training window.

Part D — Decomposition and ROI with ranges (15 pts)

9. Stacked-area decomposition of base + per-channel contributions over all 156 weeks with actual TRx overlaid; base and incremental shares in the caption.
10. Table of spend, incremental TRx, average ROI and marginal ROI (+5% in every week) per channel, ranked both ways.
11. Ranges: a moving-block bootstrap of the residuals (≥ 200 refits, blocks of 8–13 weeks) for 80% intervals on average ROI, plus the ROI implied by each channel's runner-up grid cells. One chart with both.
12. Calibration: compute the difference-in-differences lift and implied ROI from the geo holdout, and compare it with the MMM's view of the same flight. Two sentences on whether they agree and what you would do if they did not.

Part E — The reallocation memo (35 pts)

Write a two-page memo to the GLUCOREN Brand Lead. Readable by a non-technical executive; defensible to a quantitative reviewer. It must contain:

- A one-paragraph TL;DR at the top: what to cut, what to grow, the expected TRx effect with a range, and the one number you are least sure of.
- The decomposition chart as a single figure with a one-sentence caption.

- A reallocation with specific channels and dollar moves on a fixed total, in two risk postures (conservative and aggressive), with the constraint that binds first and a note on feasibility (MLR timing, contracts, hiring).
- A short list of what the model does not capture (copay program, samples, price, panel restatements, MLR cycles, anything with no variation in the data).
- One concrete experiment that would tighten your least certain estimate — design, duration, dark share, and an estimate of its cost in spend and lost TRx.

Length: two pages maximum, figures included. No code. You may cite your notebook by section. Format: PDF.

Bonus — Bayesian MMM (+10 pts)

- Fit a Bayesian MMM with PyMC-Marketing on the same data and controls. Report 80% credible intervals for average ROI per channel.
- State the posterior probability that the highest-ROI channel beats the second-highest by more than \$1 of contribution per \$ of spend.
- Optional: encode the geo-holdout result as a lift-test calibration or as an informative prior on the TV coefficient, and show what it does to the TV interval.
- Add one paragraph to the memo on whether your reallocation changes once ROI uncertainty is accounted for.

Grading rubric

Criterion	Weight	What “excellent” looks like
EDA + controls	10	Events, seasonality and ramp identified from the data; every control justified; the access-omitted comparison shown and interpreted.
Adstock + Hill	15	Correct implementations; clear plots; one-sentence interpretation per parameter; decay ranges justified from channel knowledge.
OLS MMM + diagnostics	25	AIC-selected transforms with runner-up cells; positive media coefficients; VIF discussed; hold-out MAPE reported and honestly interpreted (the formulary step).
Decomposition + ROI with ranges	15	Clean, labeled decomposition; average and marginal tables agree with the model; bootstrap and grid-sensitivity ranges on one chart; geo-holdout comparison done.
Memo to the Brand Lead	35	Clear TL;DR; reallocation specific, feasible and on a fixed total in two postures; acknowledges what the model misses; ends with a costed experiment.
Bonus — Bayesian MMM	+10	PyMC-Marketing fit converges (r-hat, divergences reported); credible intervals reported; probability statement correct; memo paragraph present.

Submission

- One zip on the course site by 11:59 PM ET on Monday, October 12: (1) Homework1_<lastname>.ipynb, (2) Homework1_Memo_<lastname>.pdf, (3) any supporting figures referenced from the memo. Do not include the data files.
- Reproducibility: the notebook must run top-to-bottom on a clean kernel with the data files in the same folder. Set numpy seed 7 at the top. Bootstrap and (if attempted) sampling should complete in under 15 minutes on a laptop.
- Late work is not accepted (syllabus). Plan for kernel crashes and for PyMC sampling time if you attempt the bonus.

AI use

You are encouraged to use AI assistants for debugging, brainstorming and explanation. The memo must be your own writing and reasoning. Include a one-paragraph “AI use” appendix at the end of the notebook naming the tool(s), what you used them for (a representative prompt is fine), and what you changed in the output. Submitting AI-generated work without disclosure is an integrity violation under the syllabus policy.

Hints (read after attempting)

- Adstock and Hill parameters are not identifiable independently of the coefficient from a single series: your search will find several cells that fit almost equally well. Pick one defensible set, report the others, and let the range show in the memo.
- A channel whose spend never varies (an always-on field force) cannot be measured well however large it is. If detailing's interval is wide, that is the finding — and the argument for a territory test.
- Saturation is why the highest-average-ROI channel is usually not the one to grow most. Always cross-check marginal ROI, and remember that marginal at +5% says nothing about +400% (try it on email).
- If your residuals show a slow wave, you under-fit the ramp or missed a control — not a channel. If they show a step at week 118, you forgot the access variable.
- The hold-out will look worse after week 118 no matter what you do. Explain why rather than hiding it.